A REPORT

ON

Intensive AI GPU Summer Internship Program

(Summer 2026)

Submitted by

Ms. Harshitha N - 20241CSE0686

Under the guidance of,

Mr. Sameer Ali
Dr. Joseph Michael Jerard V

in partial fulfillment for the award of the

degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (COMPUTER ENGINEERING)

AT

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

PRESIDENCY UNIVERSITY

BENGALURU

AUGUST 2026

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on **"Intensive AI GPU Summer Internship (Summer 2026)"** being submitted by "**Harshitha N**" bearing roll number "**20241CSE0686**" in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering (Computer Science and Engineering)** is a bonafide work carried out under our supervision.

Mr. Sameer Ali
Assistant Professor
PSCS
Presidency University

Dr. Joseph Michael Jerard V
Professor
PSCS
Presidency University

Ms. Vidhya Rengasamy
Internship coordinator
PSCS
Presidency University

Dr. Bhuvaneshwari Patil
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Dr. Nagaraja S R
Head of the Department
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Dr. Duraipandian N
Dean – PSCS & PSIS
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DECLARATION

I hereby declare that the work, which is being presented in the report entitled "**Intensive AI GPU Summer Internship (Summer 2026)**" in partial fulfillment for the award of Degree of **Bachelor of Technology** in **Computer Science and Engineering (Computer Science and Engineering),** is a record of my own investigations carried under the guidance of **Mr. Sameer Ali,** Assistant Professor and **Dr. Joseph Michael Jerard V ,** Professor, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

Harshitha N, 20241CSE0686

INTERNSHIP COMPLETION CERTIFICATE

PRESIDENCY UNIVERSITY

Presidency School of Artificial Intelligence and Advanced Computing

AI Centre of Excellence (Accelerated by NVIDIA)

Ref No: PU/PSAIAC/SIP/2026/20241CSE0686 Date: May 31, 2026

To,
Mr. / Ms. Harshitha N
Roll No: 20241CSE0686
Program / Specialization: CSE - Computer Science & Engineering (Core)

Semester: 5th Semester, Undergraduate
Presidency University, Bengaluru

Subject: Offer of Admission and Guidelines for the Intensive AI GPU Summer Internship Program (Summer 2026)

Dear Harshitha N,

Following the rigorous evaluation of your academic standing, motivation statement, we are pleased to offer you formal admission to the intensive, 2-week **AI GPU Summer Internship Program (Summer 2026)** at Presidency University. This elite training track is organized by the Presidency School of Artificial Intelligence and Advanced Computing in direct collaboration with our **NVIDIA Accelerated AI Centre of Excellence**. By accepting this offer, you will join a select cohort of students training on world-class computational architecture.

1. Individualized Training Schedule, Venue & Fee Allocations

Your program assignment contains specific technical parameters mapped to your ledger profile. You are required to attend sessions strictly within your allocated block:

Parameter	Individualized Allocation Detail
Assigned Cohort Phase	Slot 1 will be from June 1 to 12 (10 days)
Daily Time Slot Allocation	Slot A: 09:00 AM - 12:00 PM
Assigned Laboratory Venue	Lab (DGL04) - D-Block GPU Computing Lab Cluster
Registration Fee Category	Paid
Program Structural Volume	3 Hours/day over 10 Instructional Days (Total: 30 Contact Hours)
Academic Credit Load	2 Transferable Academic Credits

2. Structural Curriculum & Timeline Overview

The program is structured as a 30-hour intensive workflow incorporating 20% Lecture/Theory (6 hrs), 20% Guided Coding Workshops (6 hrs), 33% GPU Hands-on Labs (10 hrs), 20% Mentored Project Work (6 hrs), and 7% Final Assessment and Presentations (2 hrs). The 10 instructional days advance through the following core technical modules:

- **Week 1 (AI Foundations & Deep Learning Essentials):** Orientation & Environment Setup (Day 1); Machine Learning Fundamentals & Accelerated Sklearn on GPU (Day 2); Neural Networks from Scratch & PyTorch Basics on H200 (Day 3); Convolutional Neural Networks (CNNs) & GPU Profiling with Nsight (Day 4); and Hands-on Google Antigravity & Foundation Quiz (Day 5).
- **Week 2 (Advanced AI, LLMs & Capstone Project):** Transformer Architecture Deep Dive (Day 6); LLM Fine-Tuning with LoRA & PEFT via Hugging Face on H200 (Day 7); Generative AI via Stable Diffusion XL pipelines (Day 8); GPU Optimization Optimization (Mixed Precision, FlashAttention, Tensor Parallelism) & Capstone Execution (Day 9); and Capstone Project Presentations, Peer Review & Closing Ceremony (Day 10).

3. Hardware Infrastructure Utilization & AI Ethics

Throughout this internship, you will have exclusive access to a dedicated container instance on our state-of-the-art **NVIDIA H200 Tensor Core GPU Cluster** via JupyterHub. The H200 architecture is built on the Hopper (GH100) architecture, featuring 141 GB of HBM3e memory, 4.8 TB/s memory bandwidth, 3,958 FP8 Tensor TFLOPS, and integrated Transformer Engines optimized for large language model workloads.

- **Compute Provision & Suspend Policy:** GPU instances run strictly during your scheduled lab hours and will auto-suspend post-session. Temporary local container storage is wiped daily; all code must be pushed to your persistent repository before session termination.
- **Prohibited Operations:** Utilizing the H200 infrastructure for cryptocurrency mining, network scanning, unauthorized model training, or sharing infrastructure credentials will result in immediate program termination and academic suspension.
- **AI Ethics & Integrity:** While the use of generative AI tools for code layout exploration is permitted, all AI- generated code snippets must be explicitly declared in your lab logs. Plagiarism or undocumented generation will cause automatic program disqualification.

4. Mandatory Rules, Regulations, and Attendance Framework

To maintain strict institutional compliance and structural credit equivalency, you must adhere to the following framework:

- **Mandatory Attendance:** Attendance across all 10 instructional days is strictly compulsory. Any student missing more than one (1) day of instruction without prior written administrative approval from the Program Chair will be automatically disqualified and will not receive a certificate or credit.
- **Punctuality Constraint:** Late arrivals exceeding thirty (30) minutes from your designated start time (Slot A: 09:00 AM - 12:00 PM) will be marked as an absolute absence for that calendar day. Repeated late arrivals may result in immediate program removal.
- **Format Definiteness:** This is an elite, fully in-person computational lab program; no remote or hybrid participation is permitted.
- **Professional Conduct:** Professional behavior, discipline, and absolute adherence to instructions provided by mentors and coordinators are mandatory throughout the internship period.

5. Continuous Academic Assessments and Capstone Deliverables

Your performance and final credit allocation will be determined based on continuous assessment metrics weighing a total of 100 points. Successful completion requires achieving passing thresholds across all components:

1. **Daily Lab Submissions (35% Weight):** Submission of 7 core GPU lab programming modules (including CUDA Hello World, ResNet-18 Image Classification, BERT Sentiment Analysis, LLAMA 3 8B 4-bit QLoRA Fine-Tuning, SDXL Image Generation, and Nsight Profiling Sprint) at the end of each designated lab day.
2. **Week 1 Class Test / Quiz (10% Weight):** A conceptual evaluation on Core AI/ML and mathematical foundations administered on Day 5.
3. **Capstone Project (50% Cumulative Weight):** Independent execution of a chosen engineering track (Computer Vision, NLP, Generative AI, or GPU Systems Optimization) on the H200 GPU node. Final project deliverables include:
   - Fully documented, reproducible GitHub Code Repository containing functional code (20%).
   - Live Runtime Demonstration & Presentation Panel on the H200 GPU on Day 10 Afternoon (15%).
   - A 4–6 page Formal Technical Architecture Report covering problem statement, methodology, implementation, and results submitted on Day 10 Morning (15%).
4. **Peer Review & Participation (5% Weight):** Evaluated via active collaboration and peer code review cycles on Day 10.

To confirm your seat in your allotted phase, you are required to complete the final processing via your assigned payment ledger link, after which your JupyterHub credentials will be securely dispatched to your registered university email. We congratulate you on your selection and look forward to watching you build the next generation of AI systems.

AI CoE Coordination Team:
Mr. Gyanesh Verma
Mr. Vinayak Raju Kage
Mr. Thatimakula Sai Kumar
Ms. Varalakshmi T

Dr. Robin Rohit Vincent
Head, AI CoE (NVIDIA)
Presidency School of AI & Advanced Computing

STUDENT ACKNOWLEDGMENT & UNDERTAKING

I, Harshitha N, with 20241CSE0686 hereby acknowledge that I have carefully read, completely understood, and agree to strictly abide by the attendance benchmarks, strict computational infrastructure usage policies, academic integrity codes, assessment frameworks, and mandatory capstone deliverable structures (functional project, live runtime demonstration, 4-6 page technical report, and presentation slides) outlined in this offer letter. I understand that failure to comply with these stated institutional requirements will lead to immediate program disqualification, non-issuance of the Internship
Completion Certificate, and non-award of the 2 associated academic credits.

Student Signature
Harshitha N

Date & Place

AI Centre of Excellence (Accelerated by NVIDIA)

Presidency School of Artificial Intelligence and Advanced Computing

PRESIDENCY UNIVERSITY

ACKNOWLEDGEMENTS

First of all, I indebted to the **GOD ALMIGHTY** for giving me an opportunity to excel in our efforts to complete this internship on time.

I express sincere thanks to our respected Dean **Dr. Duraipandian N,** Presidency School of Computer Science and Engineering & Presidency School of Information Science, Presidency University for getting us permission to undergo the internship.

I express heartfelt gratitude to our beloved **Dr. Nagaraja S R**, Head of the Department, Presidency School of Computer Science and Engineering, Presidency University, for rendering timely help in completing this internship successfully.

I am greatly indebted to my reviewers **Mr. Sameer Ali, Assistant Professor** and **Dr. Joseph Michael Jerard V, Professor**, Presidency School of Computer Science and Engineering, Presidency University for their inspirational guidance, and valuable suggestions and for providing a chance to express technical capabilities in every respect for the completion of the internship work.

I would like to convey gratitude and heartfelt thanks to the Internship Coordinator **Dr. Bhuvaneshwari Patil** and Program Internship Coordinator **Ms. Vidhya Rengasamy.** I thank faculty and staff members of Presidency University for their support during my Internship. And also I thank my family and friends for the strong support and inspiration they have provided us in bringing out this internship.

Harshitha N (20241CSE0686)

ABSTRACT

FixNest is a web-based service management platform developed to simplify the process of finding, connecting with, and managing repair and maintenance professionals. The platform provides a centralized solution where users can explore available services, view professional profiles, communicate with service providers, make bookings, and provide ratings and feedback. Professionals can create and manage their profiles, offer their services, handle booking requests, and interact with users through the platform.

The project was developed with the aim of providing a convenient, organized, and user-friendly approach to accessing repair and maintenance services. The system integrates important functionalities such as user and professional profiles, service discovery, booking management, communication, and rating features into a single platform. The application was developed, tested, and deployed as a working web application.

Overall, FixNest demonstrates how a digital platform can simplify service discovery and improve interaction between customers and professionals. The project provides a practical solution for managing repair and maintenance requirements while offering a foundation for future enhancements such as advanced search, location-based services, online payments, improved notifications, and intelligent service recommendations.

Introduction

FixNest is a web-based service platform developed to simplify the process of finding and accessing repair and maintenance services. The platform connects users with professionals who provide different services, allowing users to explore available services, view professional profiles, communicate with professionals, and manage bookings through a single application.

The project is designed to provide a convenient alternative to traditional methods of finding service providers. It provides separate functionalities for users and professionals, allowing users to request services while enabling professionals to manage their profiles, services, and booking requests.

FixNest was developed as a practical project during the Intensive AI GPU Summer Internship. The project involved applying software development concepts to a real-world service-management problem. The completed application was tested and deployed as a working web application.

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Objective

The primary objective of FixNest is to develop a user-friendly platform that connects customers with repair and maintenance professionals in an organized manner.

The major objectives of the project are:

1. To provide a centralized platform for accessing different repair and maintenance services.
2. To allow users to discover professional service providers and view their profiles.
3. To provide a convenient booking system for requesting and managing services.
4. To enable communication between users and professionals regarding service requirements.

5. To allow users to provide ratings and feedback after receiving services.
6. To provide professionals with features to manage their profiles, services, and booking requests.
7. To develop a simple and accessible interface that provides a smooth user experience.
8. To gain practical experience in application development, testing, debugging, and deployment.

Work progress

The development of FixNest began with identifying the requirements of users and professionals and planning the overall workflow of the application. The major modules required for the platform were identified and organized before implementation.

The user interface was then developed to provide simple navigation between the different sections of the application. Separate functionalities were created for users and professionals according to their requirements.

User and professional profile management was implemented to allow users to maintain their information and professionals to present their services.

Service-related functionality was added so users could explore available repair and maintenance services and select suitable professionals.

A booking system was developed to allow users to request services and professionals to manage booking requests. Communication functionality was also implemented to allow users and professionals to exchange messages related to services and bookings.

A rating and feedback feature was included to allow users to provide feedback after receiving services. This helps improve transparency and provides useful information about service experiences.

The application was tested throughout development to identify and correct issues related to different functionalities. The individual modules were integrated and tested together to ensure that the overall workflow functioned correctly.

After development and testing, FixNest was deployed as a working web application. Deployment allowed the completed project to be accessed and tested in a live environment.

Results

The FixNest project resulted in a functional web application that provides a platform for connecting users with repair and maintenance professionals. The application successfully integrates service discovery, professional profiles, booking, communication, and rating features.

Users can access service information, explore professional profiles, request services, manage bookings, communicate with professionals, and provide ratings. Professionals can manage their profiles and services and handle booking requests from users.

The project also provided practical experience in user interface development, application functionality, data handling, testing, debugging, and deployment. The successful deployment of FixNest demonstrates the implementation of the project's core requirements in a working web environment.

Overall, the project provides a structured digital approach to managing repair and maintenance services and creates a foundation for adding further features in future versions.

Conclusion

FixNest successfully demonstrates how a web-based platform can simplify the process of connecting users with repair and maintenance professionals. The application brings important service-related functions such as professional profiles, service discovery, bookings, communication, and ratings into a single platform.

The project provided practical experience in developing and deploying a real-world-oriented application. It improved understanding of application design, feature integration, testing, debugging, and deployment.

The current system can be further enhanced in the future with features such as advanced search and filtering, location-based service discovery, online payments, improved notifications, service tracking, and intelligent service recommendations.

In conclusion, FixNest provides a practical and user-friendly solution for managing repair and maintenance service requirements and successfully achieves the main objectives of the project.

Reference

1. FixNest – Deployed web application and project implementation.
2. Official documentation of the programming languages, frameworks, and database technologies used in FixNest.
3. Official documentation and technical resources referred to during project development and deployment.
4. Internship-provided technical resources used during the development of the project.